Pop Quiz

AI-Powered Quiz Generator

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Introduction

Making self-assessment faster, personalized, and more engaging

The Learning Challenge

Students often need quick practice questions for revision, but preparing customized quizzes manually is time-consuming.

Static question banks may not match the learner's exact subject or difficulty level.

The Potential Solution

Generative AI can create relevant multiple-choice questions instantly from a user-provided subject.

This enables on-demand learning and self-assessment.

The Application

Pop Quiz provides a simple interface to enter a subject, choose the number of questions, take the quiz, and receive an instant score with explanations.

Problem Statement

Why is an AI quiz generator needed?

Traditional quiz preparation is often slow, repetitive, and not personalized to the learner's immediate needs.

Students spend time searching for suitable questions and answers.
Existing question banks may not cover a specific topic or desired difficulty.
Manual quiz creation requires significant effort from teachers and learners.
Immediate feedback and explanations are not always available.
A flexible, subject-independent quiz generation system can improve revision efficiency.

Objective

Develop a web-based AI quiz platform for instant personalized assessment

Primary Objective

To generate customized multiple-choice quizzes from any user-provided subject using generative AI and provide immediate evaluation.

Specific Goals

Accept a subject/topic from the user.
Allow the user to select the number of questions.
Generate structured MCQs using an AI model.
Provide an interactive quiz experience.
Calculate the score automatically.
Display correct answers and explanations.

Sustainable Development Goal

Contribution to quality education and lifelong learning

Goal Chosen: Quality Education

The project supports accessible, technology-enabled learning by allowing users to practice any subject on demand.

It encourages self-learning, continuous revision, and personalized assessment.

Rationale

Pop Quiz helps learners access customized practice without depending entirely on pre-created question banks.

The system can support students, educators, and lifelong learners by making self-assessment more convenient and engaging.

It aligns with the broader goal of improving access to effective learning resources.

System Workflow

From a simple topic input to an evaluated quiz

1 → 2 → 3 → 4 → 5 → 6

1. User Input

Enter a subject or topic

2. Question Request

Send structured prompt to AI

3. AI Generation

Generate MCQs in JSON format

4. Quiz Interface

Display questions and options

5. Evaluation

Compare answers with correct indices

6. Results

Show score, answers, and explanations

Data and Prompt Processing

Structured output makes AI-generated content usable by the application

Input

Subject: Python
Number of questions: 5

The application combines the user input with a carefully designed system prompt that defines:

- Question format
- Four answer options
- Correct answer index
- Short explanation
- Exact number of questions

Output

The AI is instructed to return structured JSON:

questions → question text
options → four choices
answer → zero-based correct option index
exp → short explanation

The application parses this response and renders it dynamically in the React interface.

Technology and Model Development

A modern web stack combined with generative AI

Frontend

React

Component-based UI
State management with React Hooks
Responsive interactive quiz screens

The interface includes setup, loading, quiz, and results screens.

AI Layer

Groq API

Large Language Model

Generates questions dynamically based on the selected subject.

Structured prompting is used to improve consistency and JSON parsing.

Deployment

AWS Amplify

Cloud hosting

Publicly accessible web application

Continuous deployment can be integrated with GitHub.

Security

Environment Variables

API key stored using VITE_GROQ_API_KEY during development.

Secrets should be protected and never committed to GitHub.

Model Evaluation and Application Testing

Evaluating both AI output quality and user experience

Content Quality

- Relevance to the selected subject
- Correctness of answers
- Clarity of options
- Variety in difficulty
- Quality of explanations

Application Testing

- Subject validation
- Number of questions
- API response handling
- JSON parsing
- Answer selection
- Score calculation
- Restart functionality

User Experience

- Simple subject input
- Clear progress indicators
- Interactive answer bubbles
- Immediate results
- Correct-answer explanations
- Responsive visual design

Project Impact and Effectiveness

How Pop Quiz solves the problem

Personalized Learning: Generates quizzes for almost any subject based on the user's immediate needs.
Time Efficient: Removes the need to manually search for or prepare questions.
Interactive Assessment: Provides a focused quiz experience with progress tracking.
Instant Feedback: Gives scores, correct answers, and short explanations immediately.
Scalable Solution: Can be extended with user accounts, saved quizzes, analytics, and multiple AI models.
Accessible Learning: A web-based interface can be used from any device with internet access.

Why It Will Work

The project combines personalization, automation, and accessibility

AI-Driven Personalization

The system adapts the quiz content to the user's chosen subject instead of relying on one fixed question bank.

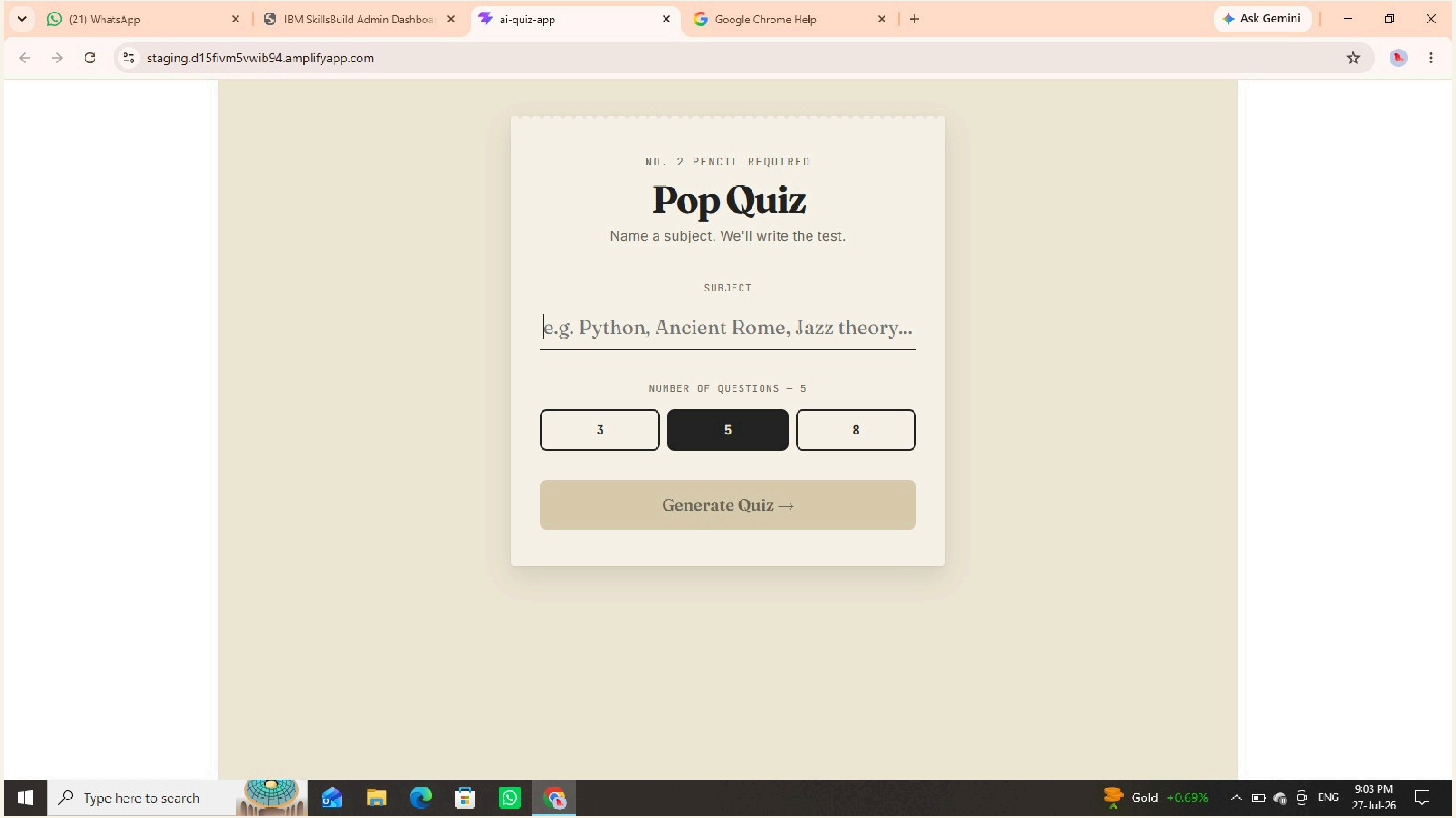
Simple User Experience

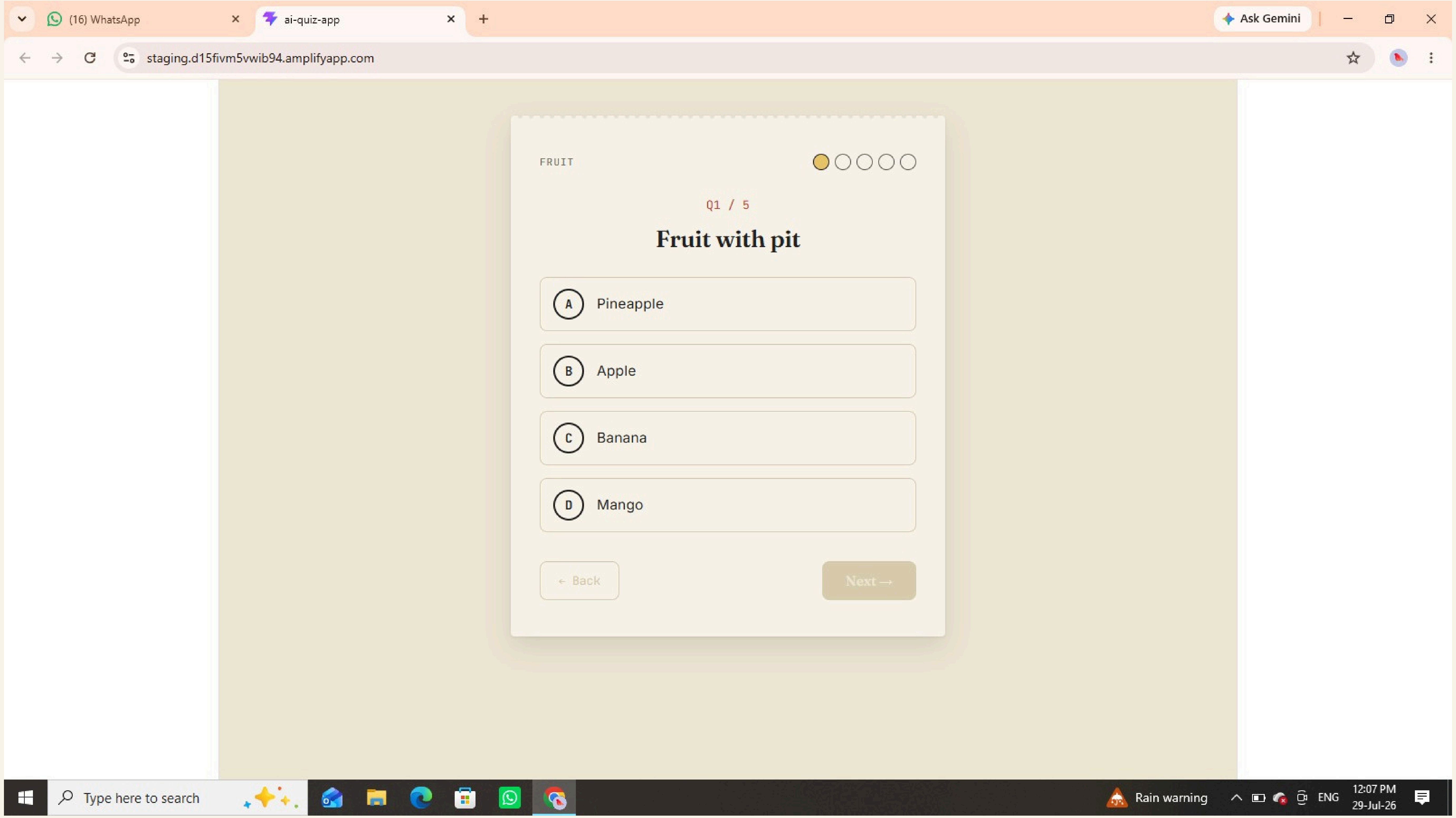
A minimal workflow—enter a subject, select question count, answer, and submit—makes the application easy to use.

Continuous Improvement

The application can evolve with better prompts, improved models, user feedback, difficulty controls, and learning analytics.

Result or Outcome

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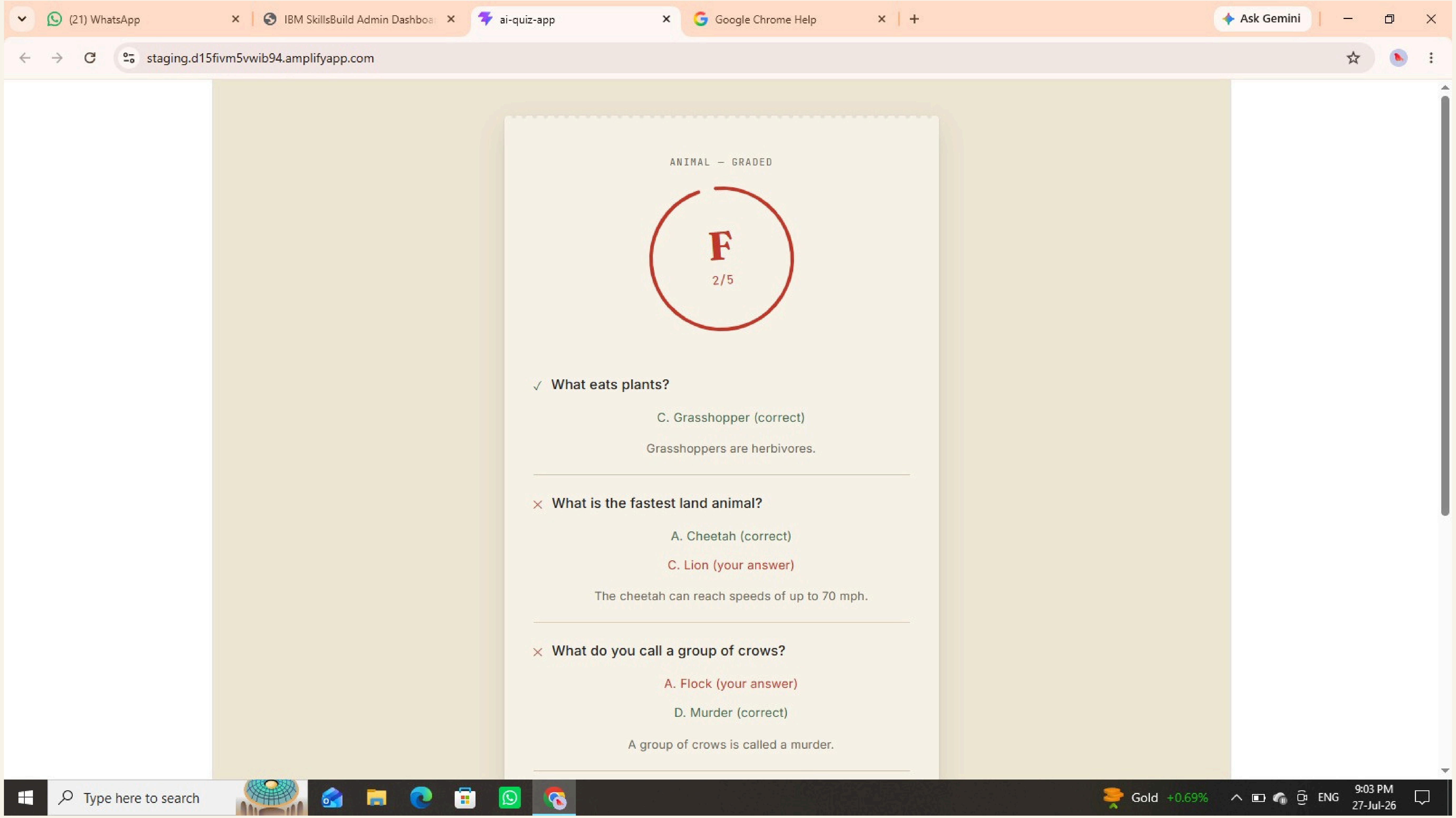
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ANIMAL — GRADED
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✓ What eats plants?
C. Grasshopper (correct)
Grasshoppers are herbivores.
× What is the fastest land animal?
A. Cheetah (correct)
C. Lion (your answer)
The cheetah can reach speeds of up to 70 mph.
× What do you call a group of crows?
A. Flock (your answer)
D. Murder (correct)
A group of crows is called a murder.
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Link to ML Model

- Deployment Id

  https://staging.d15fivm5vwib94.amplifyapp.com/

Conclusion

Pop Quiz demonstrates how generative AI can transform traditional self-assessment into a personalized, interactive, and accessible learning experience.

The project successfully combines React, an AI-powered API, structured prompting, dynamic JSON processing, and cloud deployment to create a practical educational application.

Thank You